Research Article

Coach WAR: A Counterfactual Approach to Evaluating NFL Head Coach Impact in the Modern Era

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Abstract: Evaluating head coach performance in the National Football League presents unique challenges because of the complex interactions among coaching decisions, player talent, and organizational factors. We introduce a novel counterfactual framework for measuring head coach impact by predicting how each team would have performed under a replacement-level coach with league-median characteristics, then comparing this baseline to actual results. This difference represents coaching Wins Above Replacement (WAR), the wins directly attributable to that coach's performance. Analyzing 1,637 team-seasons from 1970 to 2024 with 365 features spanning draft history, injuries, rosters, team performance, and coaching histories, we employ XGBoost regression to isolate coaching excellence from roster quality and organizational context. The model achieves $\mathrm{R}^2 = 0.815$ on the full dataset, with coaching-specific features accounting for 44% of total model importance. Results reveal substantial coaching impact, with elite coaches averaging up to +2.4 WAR per season and poor coaches averaging up to -3.2 WAR. Analysis of offensive versus defensive coaching backgrounds reveals a dramatic historical shift: defensive coaches dominated the 1970s (+0.80 win advantage, p=0.007), but offensive coaches now hold a +0.58 win edge in the 2020s (p=0.10). Year-to-year persistence analysis demonstrates scheme obsolescence, with defensive coaches showing significantly stronger multi-year persistence (2-year: R^2=0.180 vs 0.069, p=0.009). Our framework provides NFL organizations with objective, context-adjusted coaching evaluation comparable to analytical rigor applied to player personnel decisions.

Keywords: coaching analytics, wins above replacement, NFL, sports analytics

1 Introduction

Evaluating head coach performance in the National Football League presents unique challenges because of the complex interactions among coaching decisions, player talent, and organizational factors. Traditional metrics often conflate team success with coaching quality, failing to isolate a coach's actual contribution to the team.

We introduce a novel counterfactual framework for measuring head coach impact: we predict how each team would have performed under a replacement-level coach with league-median characteristics, then compare this baseline to actual results. This difference represents coaching Wins Above Replacement (WAR), the wins directly attributable to that coach's performance. A higher WAR indicates stronger coaching performance, and conversely, a lower WAR indicates poorer coaching performance. By controlling for roster quality, strength of schedule, and organizational resources, we isolate coaching excellence or deficiency from the talent and context surrounding it.

2 Literature Review

Prior research on coaching evaluation has taken several approaches. Hadley et al. (2000) used a Poisson regression framework with basic team statistics to assess NFL coaching efficiency relative to production frontiers, while Goff and Wisley (2006) examined managerial lifecycle effects. This work expands on Hadley et al. (2000) substantially by isolating coaching impact from characteristics beyond roster quality, including salary distribution, player turnover, actual injury data, and extensive coach-specific performance characteristics. Our approach yields actionable insights for contemporary NFL decision-making while Hadley et al. (2000)'s work established coaching as a significant production factor.

More recently, sports analytics has embraced WAR methodologies for player evaluation in baseball (Palmer and Thorn, 1984; Smith, 2008), basketball (Berri et al., 2006), and other sports. Our contribution extends this framework to coaching evaluation, adapting the counterfactual approach to handle the unique challenges of isolating coaching impact in a team sport where coaches don't directly execute plays.

3 Methods

We analyzed 1,637 team-seasons spanning 1970–2024, representing all 32 current NFL franchises. This period corresponds to the Modern Era of professional football. Our analysis incorporated 365 features spanning multiple data sources. Data included draft pick history, injury reports, starter designations, complete rosters, historical team performance, and comprehensive head coach career histories from Pro Football Reference, along with salary cap allocation and contract details from Spotrac. Feature counts by category are shown in Appendix A (Table 7). The coach-specific feature set comprised 139 features capturing coaching tenure and previous team performance during previous hiring instances as either an offensive coordinator, a defensive coordinator, or a head coach. Data availability varied by data category. While team performance and roster metrics extended across all seasons, salary cap data was limited to 2011–2024, and player injury data was limited to 2009–2024.

3.1 The Counterfactual Framework

To isolate coaching impact from roster quality, organizational context, and other applicable factors, we developed a machine-learning-based counterfactual approach to answer the question: what would this team's performance have been if we had replaced the head coach with a league-median alternative while keeping everything else identical?

Specifically, we define replacement-level coaching as a coach with league-median characteristics across all 139 coach-specific features, including career tenure and historical performance metrics. When generating the replacement-level prediction, we substitute the actual coach's features with these league-median values while holding all other factors constant. This isolates what a replacement coach would be expected to achieve given identical circumstances. Coach WAR is then calculated as the difference between the team's actual winning percentage and this replacement-level prediction.

This approach parallels established WAR methodologies in baseball and basketball (Berri et al., 2006; Palmer and Thorn, 1984; Smith, 2008; Woolner, 2002), where player contributions are measured against readily available replacement-level alternatives. Unlike baseball's replacement level (set at a .294 winning percentage, representing the performance of readily available minor league or free agent players (Appelman and Cameron, 2013; Baseball-Reference.com, n.d.)), we define replacement-level coaching as the league median. This choice reflects the coaching labor market's unique structure: NFL head coaches average only 3.2 years of tenure (Goff and Wisley, 2006), substantially shorter than MLB players' average career of 5.6 years (Witnauer et al., 2007). Thus, this Coach WAR measures performance relative to the average NFL head coach rather than a theoretical minimum-quality hire.

3.2 Model Training

We employed XGBoost regression (Chen and Guestrin, 2016) to predict team winning percentage with coach median metrics, selected for its ability to capture complex non-linear relationships and feature interactions while providing robust feature importance metrics. The model was trained on all 365 predictor features (excluding the target variable Win_Pct and metadata columns Team and Year) with winning percentage as the target. To handle missing values in the dataset, primarily arising from incomplete early-era salary data and injury information, we used Singular Value Decomposition (SVD) matrix completion (Brand, 2002), which preserved data structure while imputing missing values based on observed patterns across similar team-years.

To minimize potential overfitting, we employed a coach-based cross-validation strategy, ensuring coaches in the test set were entirely unseen during training (81.7% train, 18.3% test). Hyperparameters were selected through randomized search with 50 iterations and 5-fold cross-validation. Final hyperparameter values are shown in Appendix C (Table 8).

3.3 WAR Calculation

The XGBoost model predicts the average winning percentage (Win_Pct) for each team under the same circumstances with a league median coach. WAR is calculated according to the following equation:

$$\mathrm{WAR} = 16 \times (\mathrm{Win\_Pct}_{\mathrm{actual}} - \mathrm{Win\_Pct}_{\mathrm{predicted}}) \tag{1}$$

For historical comparability, we standardize all WAR calculations to a 16-game season, even for the 17-game seasons (2021+). This ensures coaches from different eras are evaluated on an equivalent basis and prevents artificial inflation of recent WAR values. For 17-game seasons, this represents a 6% conservative adjustment in absolute measurement.

4 Results

The final model achieved an $\mathrm{R}^2 = 0.815$ on the full dataset, demonstrating strong explanatory power for team performance. To validate model generalization, we used a coach-based holdout strategy, randomly withholding ∼20% of coaches (54 of 270) along with all their seasons from training. The model achieved $\mathrm{R}^2 = 0.615$ on these completely unseen coaches (gap = 0.248 vs. training set). While this gap indicates overfitting, the test performance remains meaningful because: (1) regularization (max depth = 3, ℓ_1/ℓ_2 penalties) prevented more severe overfitting, (2) tree-based ensemble models inherently trade some generalization for pattern-detection capability, and (3) our coaching WAR analysis relies on *relative* comparisons between actual and replacement-level coaching rather than absolute win percentage predictions. To verify our findings are not artifacts of the XGBoost algorithm, we compared results to Ridge regression (linear baseline). Coach WAR estimates from both approaches correlated highly (Spearman $\rho = 0.85$), confirming relative rankings are robust to modeling approach. XGBoost achieved superior predictive performance (test $\mathrm{R}^2 = 0.615$ vs. 0.543 for Ridge), justifying its selection for the primary analysis. Additionally, the mean coaching WAR across all team-seasons was 0.01 games (median = 0.02), validating that the median replacement baseline correctly centers the distribution near zero. Feature importance analysis confirmed coaching-specific features accounted for 44% of total model importance, demonstrating that coaches exert substantial influence on outcomes when controlling for roster quality, financial resources, and competitive context. Appendix D (Table 9) shows the top 20 features by importance.

4.1 Football Context and Career Leaders

While single-season WAR ranged from +7.1 to -8.4 (as discussed in section 3.7), career averages provide a more robust assessment of coaching ability across varying contexts. Among coaches with at least 3 seasons coached, the top performers averaged between +1.4 and +2.6 WAR per season. Conversely, the worst coaches with at least 3 seasons ranged from -3.1 to -1.9 WAR per season. Tables 1 and 2 show the best and worst coaches, respectively, by average WAR per season among coaches with at least 3 seasons coached.

Tab. 1: Top 15 Coaches by Average WAR per Season (minimum 3 seasons)

Rank	Coach	Seasons	Total WAR	Avg. Yearly WAR
1	Kevin O'Connell	3	+7.1	+2.4
2	Don Shula	26	+50.6	+1.9
3	Mike Martz	6	+11.0	+1.8
4	John Madden	9	+15.0	+1.7
5	Mike Vrabel	6	+9.4	+1.6
6	John Ralston	5	+7.5	+1.5
7	Bruce Arians	8	+11.3	+1.4
8	Raymond Berry	5	+7.0	+1.4
9	Mike McDaniel	3	+4.2	+1.4
10	Red Miller	4	+5.4	+1.3
11	Bill Belichick	29	+37.3	+1.3
12	Chuck Pagano	6	+7.5	+1.3
13	Kyle Shanahan	8	+9.8	+1.2
14	Joe Gibbs	16	+19.2	+1.2
15	Tom Landry	19	+22.3	+1.2

Tab. 2: Bottom 15 Coaches by Average WAR per Season (minimum 3 seasons)

Rank	Coach	Seasons	Total WAR	Avg. Yearly WAR
1	Matt Eberflus	3	-9.6	-3.2
2	Pat Shurmur	4	-12.2	-3.1
3	Hue Jackson	4	-12.1	-3.0
4	J.D. Roberts	3	-8.2	-2.7
5	Scott Linehan	3	-8.0	-2.7
6	Jack Patera	7	-18.4	-2.6
7	Kay Stephenson	3	-7.5	-2.5
8	Bruce Coslet	8	-18.7	-2.3
9	Bill Arnsparger	3	-7.0	-2.3
10	David Shula	5	-11.5	-2.3
11	Joe Bugel	5	-10.4	-2.1
12	Rod Marinelli	3	-5.9	-2.0
13	Paul Wiggin	3	-5.8	-1.9
14	June Jones	3	-5.6	-1.9
15	Ed Biles	3	-5.5	-1.8

The reigning coach of the year, Kevin O'Connell, currently leads all coaches with an average WAR of +2.4 per season across three seasons with Minnesota (2022–2024). Don Shula (+1.9 wins per season over 26 seasons) and Mike Martz (+1.8 wins per season over 6 seasons) round out the top three.

The career leader board validates the framework's ability to identify coaching excellence across eras. Legendary coaches Don Shula, John Madden, and Tom Landry appear prominently, with Shula averaging

+1.9 WAR per season across an unprecedented 26 seasons (total career WAR of +50.6) and Landry posting +1.2 WAR per season across 19 seasons (total +22.3 WAR). These sustained performances across coaching legends demonstrate the validity of this approach in calculating coach WAR.

At the opposite extreme, the poorest-performing coaches demonstrated persistent struggles across multiple seasons. Matt Eberflus recorded the lowest average at -3.2 WAR per season across 3 seasons (total -9.6 WAR). Pat Shurmur's -3.1 WAR per season ranked the second-worst all-time.

The spread between elite and poor career performance is substantial: approximately 5.5 wins per season separates coaches at the 95th percentile (roughly +2.0 wins per season) from those at the 5th percentile (roughly -2.5 wins per season). For NFL franchises, this represents the vital difference between perennial playoff contention and chronic rebuilding.

4.2 Coach WAR in Aggregate

Figure 1 illustrates the relationship between coaching longevity and WAR performance by plotting coaches based on average WAR per season and career length (measured in seasons coached).

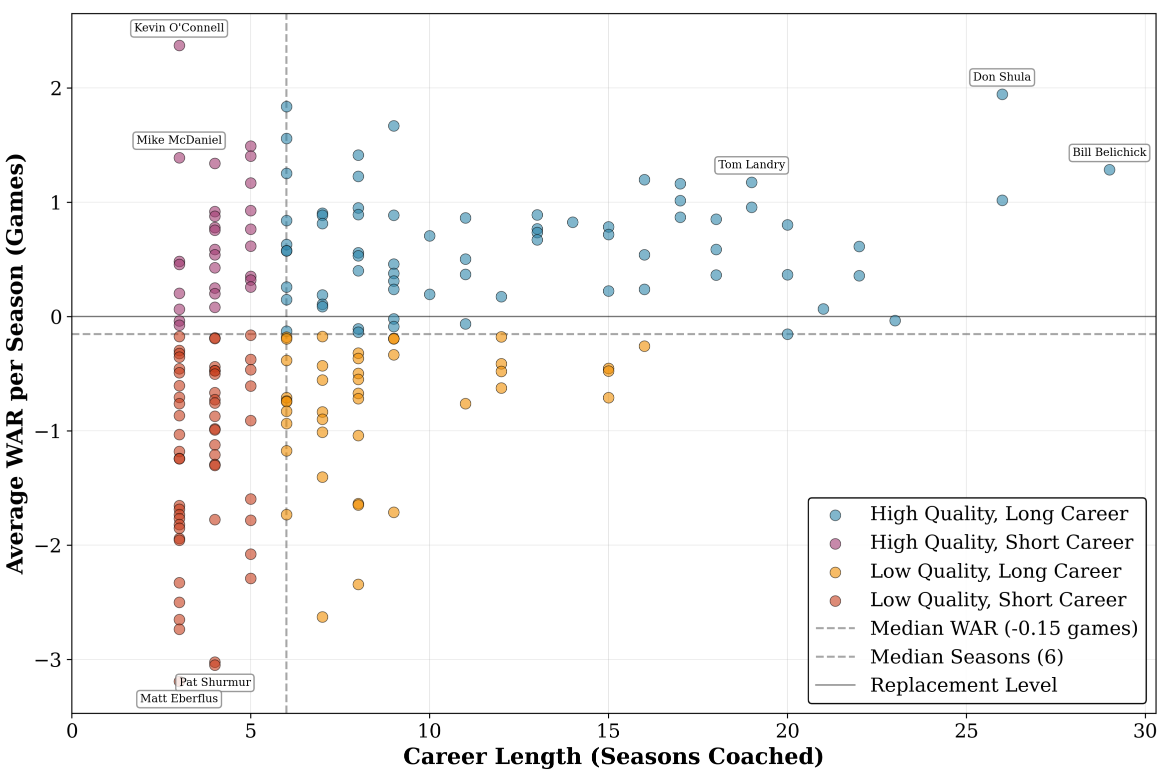

Fig. 1: Coach WAR Career Distributions

While the quadrant breakout based on medians is largely illustrative, a few takeaways become clear. The upper-right and lower left quadrants show a clear positive relationship between coaching quality and career length. While this should be no surprise—since it makes sense for successful coaches to be retained—it does validate market efficiency. However, the number of coaches with sustained careers of 5+ seasons despite averaging near-zero or negative WAR (lower-right quadrant) represents a market inefficiency, as organizations historically retained coaches who demonstrably cost wins. Isolating these long-tenured, poor-performing coaches for potential replacement is a primary use case for this Coach WAR analysis.

4.3 Current Coach Analysis

Figure 2 shows the average career WAR per season for 2024 coaches against their career length. Blue markers indicate coaches who were retained into the 2025 season, and red markers indicate coaches who were fired.

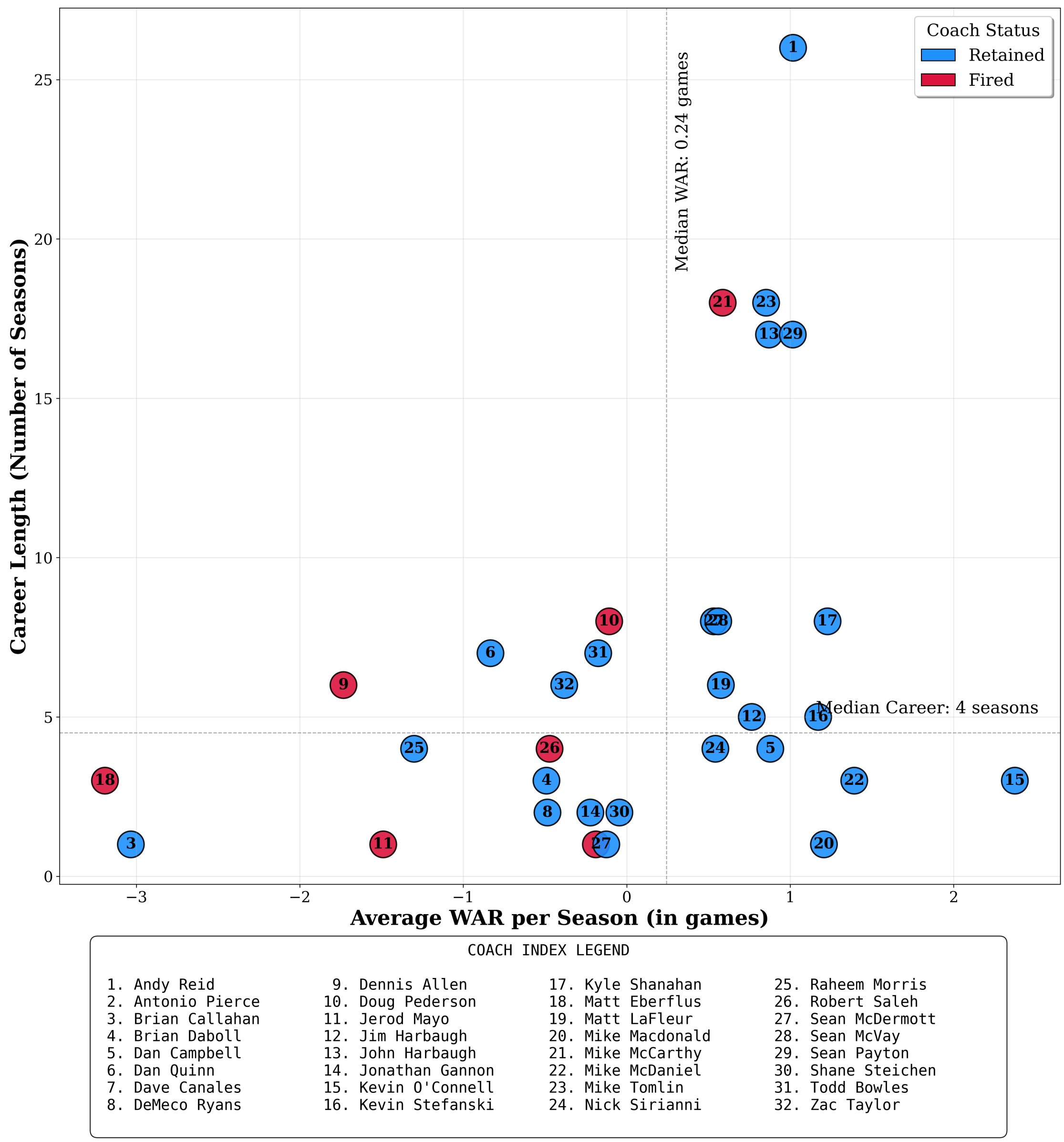

Fig. 2: 2024 Head Coach Average WAR per Season versus Career Length

These averages, however, are not the whole story. To dig deeper into the firing decisions of coaches, it's necessary to isolate their performance in the 2024 season. Figure 3 below shows the cumulative WAR trajectories for these coaches, and Figure 4 below highlights each coach's 2024 WAR.

Figure 4 shows how much the firing of Mike McCarthy stands out from that of other fired coaches. With a 2024 WAR of +1.3, Mike McCarthy ranked in the top 30% of NFL head coaches in the season. In

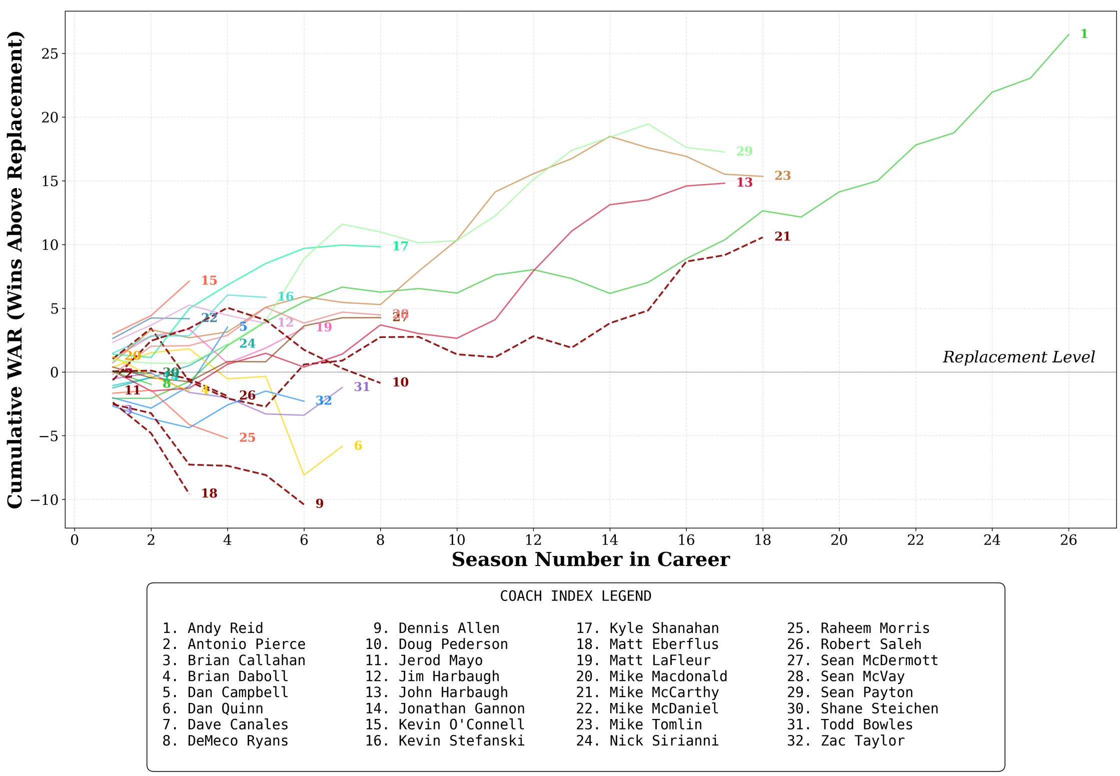

Fig. 3: Coach WAR Trajectory for 2024 Head Coaches

contrast, three of the seven fired coaches were in the bottom 4 for 2024 Coach WAR. Brian Callahan was the lone exception (a bottom 4 coach who was not fired), but he has since been fired in the 2025 season[1].

4.4 Offensive versus Defensive Coaches

One debate that will likely never be satisfied is whether offensive or defensive coaches are more successful. For this analysis, coaches were classified as offensive or defensive based on their experience as coordinators or position coaches prior to becoming a head coach. Offensive backgrounds included offensive coordinators and offensive position coach roles (quarterbacks, wide receivers, etc.), and defensive backgrounds included defensive coordinators and defensive position coach roles (linebackers, secondary, etc.). We analyzed Coach WAR by coach type (offensive, defensive, or other/indeterminate) to identify whether coaches of a specific background outperformed other coaches. This data provides evidence that, at least through the lens of circumstance-adjusted performance, defensive-minded head coaches hold a slight edge ($p=0.056$) on average in the modern era (1970+), though the effect size is negligible (Cohen's d = -0.070). But how does the evolution of the game of football play into this analysis? To answer this question, this project segmented the data by decade to isolate the average coach impact for each coach type over time. The results are shown in Table 3 below. Mann-Whitney U-Test results are shown in Table 4 for each decade.

The NFL's evolution from a defense-dominated to offense-dominated league is clearly reflected in coaching success patterns across eras. In the 1970s, defensive-minded head coaches averaged +0.29 WAR per season compared to -0.51 for offensive coaches, a +0.80-win gap favoring defensive backgrounds ($p=0.007$).

1 Analysis conducted October 2025 on 1970–2024 data; the partially complete 2025 season was excluded from this analysis

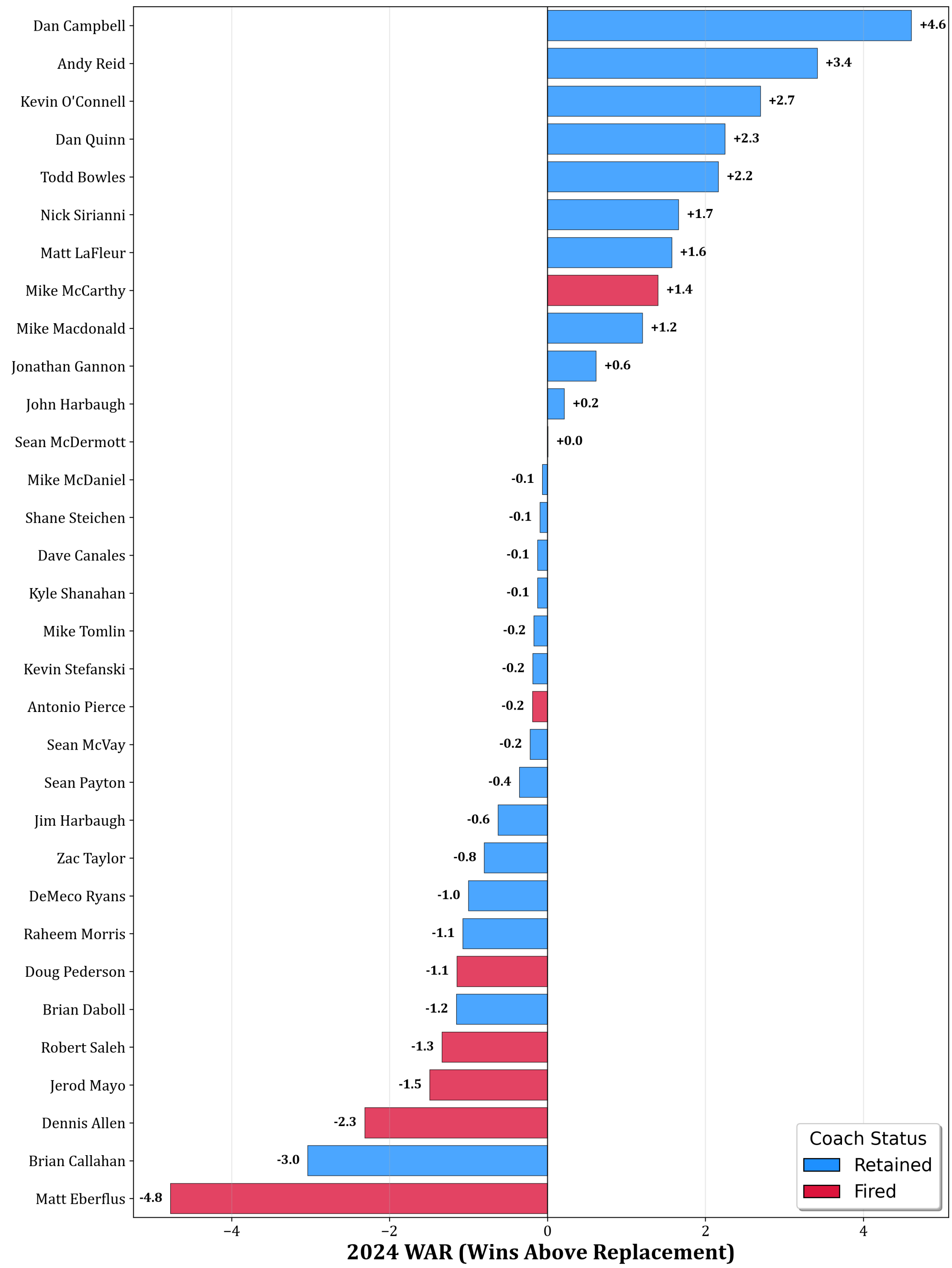

Fig. 4: 2024 Coach WAR for 2024 Head Coaches

By the 2000s, the two groups were essentially even (+0.01 vs. -0.01, p=0.954), and in the 2020s, offensive coaches have established a marginally significant edge, averaging +0.28 WAR per season compared to just -0.30 for defensive coaches (p=0.10). It should be noted that if the trend between offensive and defensive coaches holds for the remainder of the 2020's, that the difference in WAR would be statistically significant with the larger sample size.

The cumulative shift is clear: defensive coaches have declined by 0.59 WAR per season from the 1970s to 2020s, while offensive coaches have improved by 0.79 WAR per season over the same period, a complete reversal in relative performance. In today's NFL, offensive expertise appears to provide a competitive advantage that didn't exist in previous generations.

Tab. 3: Average Coach WAR per Season by Coach Background and Decade

Background	1970's	1980's	1990's	2000's	2010's	2020's
Offensive	-0.51	+0.29	-0.37	+0.01	+0.01	+0.28
Defensive	+0.29	+0.01	+0.27	-0.01	+0.09	-0.30
Other	-0.07	+0.00	+0.63	—	—	—
Delta, 2020's vs 1970's						
Offensive	+0.79					
Defensive	-0.59					

Tab. 4: Mann-Whitney U Test comparing Coaching Backgrounds by Decade

Metric	1970's	1980's	1990's	2000's	2010's	2020's
Sample Size (O/D)	34/21	30/22	33/28	40/37	42/33	33/20
Seasons (O/D)	87/82	146/83	143/126	152/151	158/135	92/56
Mean WAR (O/D)	-0.511/+0.290	+0.293/+0.015	-0.371/+0.267	+0.011/-0.008	+0.012/+0.092	+0.275/-0.300
Difference (O - D)	-0.801	+0.278	-0.638	+0.019	-0.080	+0.575
U-statistic	2711	6389	7358	11415	10449	2993
p-value	0.007	0.494	0.005	0.954	0.710	0.100
Significance	Highly Sig.	—	Highly Sig.	—	—	Marginally Sig.

4.5 WAR Persistence and Scheme Obsolescence

Coaching performance is unstable from year to year. Analyzing 1,163 consecutive season pairs from 1970–2023, Year N coaching WAR explains only 7.3% of the variance in Year N+1 performance (Figure 5). Elite coaches averaging +2.75 wins above replacement the following season regress dramatically to just +0.68, losing ∼75% of their advantage. In contrast, the worst-performing coaches at -2.59 WAR improve substantially to -1.29 WAR when implementing a 10th percentile penalty for being terminated (Figures 6).

This volatility contrasts sharply with raw winning percentage, which shows nearly double the year-to-year stability ($R^2 = 0.137$, Appendix E, Figure 7). Winning percentage likely captures persistent organizational advantages, such as market size, ownership stability, accumulated roster talent, that carry forward regardless of coaching. The coach-specific contribution, by contrast, fluctuates, as what worked last season often doesn't work this season. This represents empirical evidence for scheme obsolescence.

But the instability isn't uniform across all coaches. While both offensive and defensive coaches show similar single-year persistence ($R^2 = 0.064$ and 0.092 respectively), multi-year averaging exposes a stark divergence. When evaluating coaches based on 2-year average performance, defensive coaches' predictive power approximately doubles to $R^2 = 0.180$. Offensive coaches' predictive power remains relatively flat at $R^2 = 0.069$. This is a statistically significant difference at 99% confidence (Fisher's Z p=0.009, Appendix F, Figure 8 and Table 10). At three-year horizons, the gap persists (p=0.020). The difference in persistence between Offensive and Defensive coaches highlights that Offensive schemes may have a shorter shelf life in the modern NFL. These patterns suggest fundamentally different retention strategies by coordinator background. Offensive coaches coming off elite seasons should trigger caution, not contract extensions, as their advantage will likely erode as schemes become known.

4.6 Limitations and the "Franchise Quarterback Effect"

Tables 5 and 6 show the top and bottom 10 single-season WAR instances in the modern era.

The highest single-season coaching WAR values reveal an important limitation of roster-adjusted evaluation. In these extreme performances, two of the top 10 single-season WAR values belong to coaches during years with elite quarterback play (Peyton Manning in 2009, and Drew Brees in 2011). This "franchise

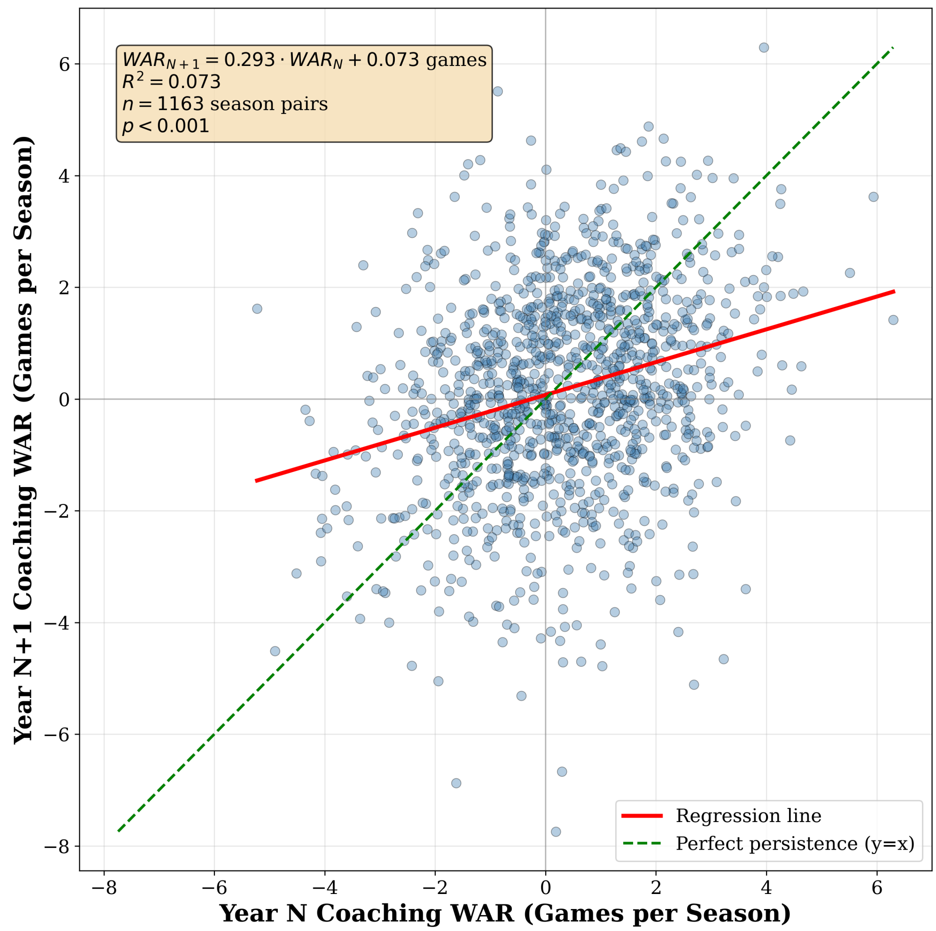

Fig. 5: Coach WAR in Year N+1 vs. Year N (persistence)

Tab. 5: Top 10 Single-Season WAR Instances

Rank	Team	Year	Coach	Actual Win %	Replacement Win %	WAR
1	CHI	1987	Mike Ditka	0.733	0.340	+6.3
2	CLT	2009	Jim Caldwell	0.875	0.504	+5.9
3	WAS	1982	Joe Gibbs	0.889	0.545	+5.5
4	NOR	2011	Sean Payton	0.812	0.508	+4.9
5	NWE	2007	Bill Belichick	1.000	0.708	+4.7
6	NYG	1986	Bill Parcells	0.875	0.586	+4.6
7	DET	2024	Dan Campbell	0.882	0.594	+4.6
8	RAI	1976	John Madden	0.929	0.648	+4.5
9	MIA	1993	Don Shula	0.562	0.284	+4.4
10	SFO	1987	Bill Walsh	0.867	0.590	+4.4

quarterback effect" represents an inherent challenge in isolating coaching contributions: elite quarterbacks and elite coaching are rarely independent. While our framework controls for roster quality through metrics like player age, experience, and positional spending, it cannot fully disentangle whether a coach maximizes quarterback talent or whether an elite quarterback makes ordinary coaching appear exceptional. However, our framework's ability to identify coaching excellence in the absence of elite QB play (Dan Campbell 2024, defense-focused teams in the 1970s and 1980s) suggests it captures more than just QB quality. Moreover, career-average analysis naturally controls for this.

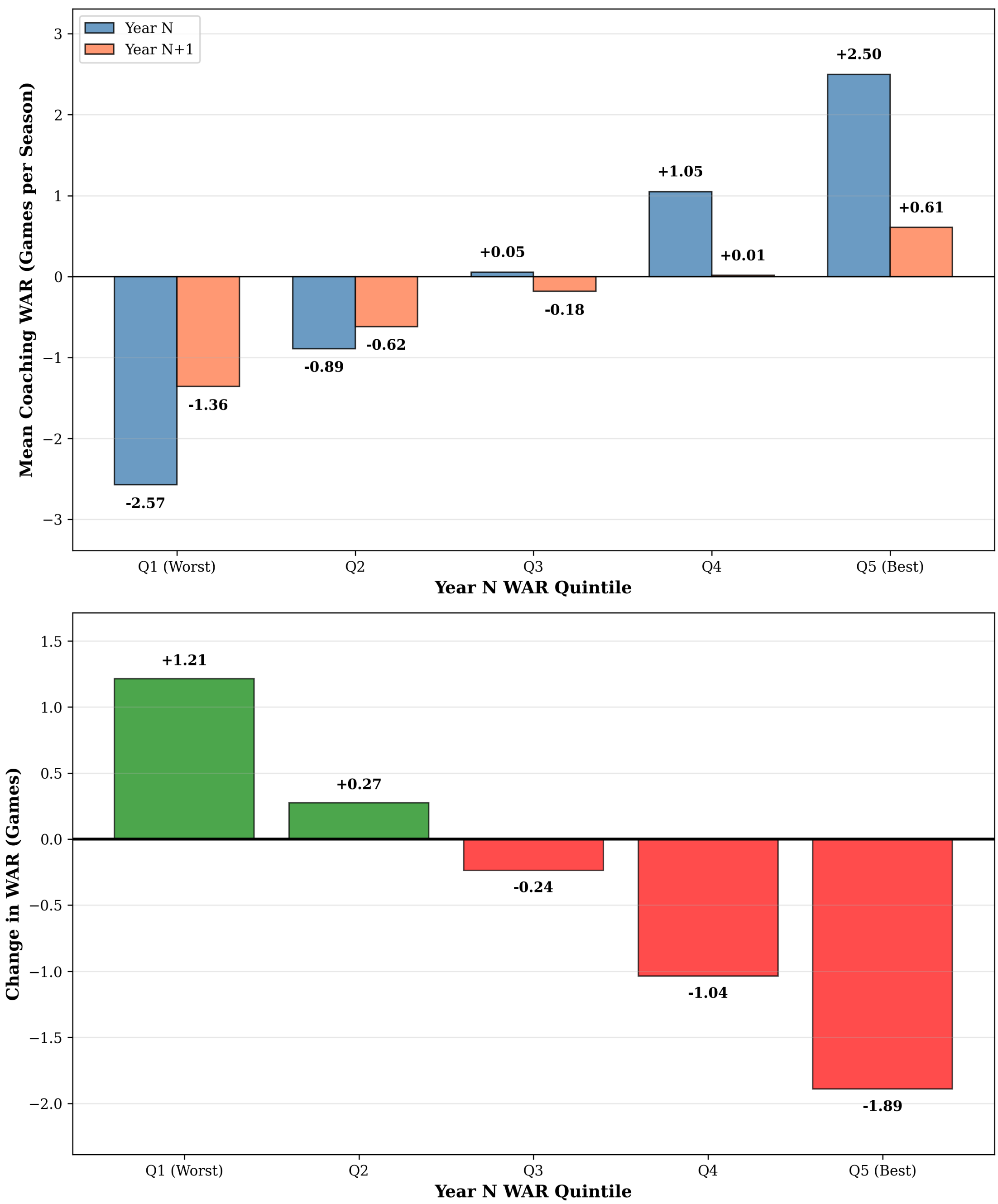

Fig. 6: Mean Coach WAR in Year N and Year N+1 by Quintile

At the opposite end of the spectrum, the poorest single-season performances highlight many coaches who were winless prior to being fired in that season. In our approach, we still multiply partially coached seasons by 16 games to estimate the full-season impact. This overstates the negative impact of being fired midseason, which to be fair, is a significant representation of performance.

The ∼14 win spread between the best and worst single-season performances highlights coaching's potential impact, even as the franchise quarterback effect reminds us that extreme values should be interpreted with appropriate caution.

Additional methodological limitations warrant consideration. First, while the model controls for roster quality through age, experience, turnover, and salary allocation, these proxy measures cannot fully capture

Tab. 6: Bottom 10 Single-Season WAR Instances

Rank	Team	Year	Coach	Actual Win %	Replacement Win %	WAR
1	ATL	2020	Dan Quinn	0.000	0.484	-7.7
2	PHI	1971	Jerry Williams	0.000	0.438	-7.0
3	SEA	1982	Jack Patera	0.000	0.430	-6.9
4	CIN	1996	David Shula	0.143	0.560	-6.7
5	DET	1976	Rick Forzano	0.250	0.582	-5.3
6	SFO	1979	Bill Walsh	0.125	0.451	-5.2
7	HTX	2020	Bill O'Brien	0.000	0.320	-5.1
8	RAM	2008	Scott Linehan	0.000	0.315	-5.0
9	CLE	2016	Hue Jackson	0.062	0.369	-4.9
10	WAS	2019	Jay Gruden	0.000	0.299	-4.8

player talent. Teams with similar demographic profiles may contain vastly different talent levels. Second, organizational factors beyond our data, such as ownership stability, drafting competence, and facility quality, likely influence outcomes but remain unobserved.

4.7 Practical Applications

The Coach WAR framework offers immediate practical value for NFL front offices facing high-stakes personnel decisions. In hiring decisions, rather than relying on win-loss records that conflate coaching with roster quality, teams can evaluate candidates based on circumstance-adjusted performance. A coach with a .500 record but +1.5 WAR per season likely outperformed a coach with a .600 record but -0.5 WAR who inherited superior talent. Similarly, contract negotiations benefit from quantifiable performance metrics for compensation discussions, analogous to how player WAR informs salary arbitration in baseball. These applications provide decision-makers with objective baselines for evaluating coaching contributions independent of the circumstances coaches inherit.

Finally, retention decisions benefit from objective performance baselines. Organizations can identify underperforming coaches earlier and avoid prolonged rebuilds. Matt Eberflus's cumulative -6.4 WAR through two seasons provided quantitative evidence that performance issues were structural rather than circumstantial, validated by his subsequent -3.2 single-season WAR in 2024. Conversely, coaches like Mike McCarthy (+1.3 WAR in 2024 despite dismissal) demonstrate that firing decisions based on win-loss records alone may discard above-replacement talent.

5 Conclusion

Head coaching represents one of the most consequential investments in professional sports. In this project, we quantify what NFL decision-makers have long struggled to measure: coaching quality, which fundamentally alters franchise trajectories independent of roster talent.

The practical implications are immediate and substantial. Front offices can now evaluate coaching candidates and incumbent coaches with circumstance-adjusted metrics that separate coaching impact from inherited advantages. Kevin O'Connell's +2.4 WAR per season with Minnesota demonstrates impressive coaching performance despite a roster that hasn't been considered exceptional. Conversely, Matt Eberflus's year 2 cumulative WAR of -6.4 could have prompted the organization to move on and avoid a lame-duck situation for Caleb Williams' rookie year.

The evolution of coaching value by coordinator background tells an equally compelling story about the modern NFL. Defensive-minded coaches dominated from the 1970s through the 1990s, but offensive coaches now generate 0.58 more WAR per season in the 2020s ($p=0.10$). Organizations hiring head coaches today should weigh offensive expertise more heavily than historical precedent would suggest, as the game itself has fundamentally shifted. Yet year-to-year performance analysis reveals this offensive advantage erodes rapidly: while defensive coaches' predictive power more than doubles when evaluated over multiple seasons, offensive coaches' predictive power remains flat over the same timeframe. This asymmetry suggests offensive schemes depreciate while defensive skills compound, a market dynamic with direct implications for retention decisions.

As NFL teams increasingly embrace analytics for player evaluation, from Expected Points Added to catch probability models, coaching assessment has remained surprisingly subjective, relying on win-loss records that confound coaching quality with roster talent, or playoff success that introduces large small-sample noise. But the stakes are too high for subjective evaluation: a single coaching decision can swing a franchise's trajectory by 5+ wins per season, on average, and by $\sim$14 wins in extreme circumstances.

The key question facing NFL organizations is whether they will continue to make eight-figure coaching decisions based on intuition and non-context-adjusted performance, or whether they will embrace objective, context-adjusted evaluation that separates true coaching impact from the circumstances coaches inherit. This Coach WAR framework provides the objective baseline the industry needs, quantifying the wins directly attributable to coaching excellence or deficiency, isolated from roster advantages, schedule strength, and organizational resources. Front offices armed with circumstance-adjusted coaching metrics can make these critical decisions with the same analytical rigor they now apply to draft picks and free agent signings, and the franchises that adopt it first will gain a sustainable competitive advantage in the league's most impactful personnel decision.

Data Availability

Coaching history data was scraped from Pro Football Reference. Processed data and analysis code are available at https://github.com/jwilliamson7/Coach_WAR or upon request to the corresponding author. Raw salary cap data from Spotrac is proprietary but processing scripts are provided.

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A Feature Count Across Categories

Tab. 7: Feature Count Across Categories

Feature Category	**Coach Feature?**	**Number of Features**
Metadata (excluded from training)	No	2
Salary Cap	No	19
Roster Turnover	No	63
Starter Turnover	No	27
Injury/Availability	No	37
Player Demographics	No	44
Penalty/Opponent	No	7
Draft Capital	No	29
Coaching Experience	Yes	7
Prior Team Performance (OC)	Yes	33
Prior Team Performance (DC)	Yes	33
Prior Team Performance (HC)	Yes	66
Target Variable	No	1
Total Columns in Dataset		**368**
Features Used in Training		**365**

B Selected Hyperparameters for the Final Model

Tab. 8: Selected Hyperparameters for the Final Model

Hyperparameter	**Value**
Objective	reg:squarederror
Random State	42
Number of Estimators	250
Learning Rate	0.03
Max Estimator Depth	3
Gamma	0
Lambda	0.1
Alpha	0.1
Subsample	0.6
Colsample by Tree	0.6
Minimum Child Weight	6

C Feature Importance Ranking of Top 20 Features

Tab. 9: Feature Importance Ranking of Top 20 Features

Rank	Feature Description	Importance	Coach Metric
1	Reserve/injured/suspended cap as percentage	0.041	No
2	Normalized opponent interceptions thrown	0.025	No
3	Normalized opponent points scored under HC	0.025	Yes
4	Normalized team interceptions thrown	0.025	No
5	Normalized opponent yards allowed under HC	0.023	Yes
6	Average age of entire roster	0.016	No
7	Active 53-man roster cap as percentage	0.016	No
8	Normalized passing TDs under HC	0.016	Yes
9	Average percentage of games missed by QBs	0.016	No
10	Number of new QBs added to roster	0.014	No
11	Average experience of offensive linemen	0.012	No
12	Normalized 4th down attempts under OC	0.010	Yes
13	Normalized points scored under HC	0.010	Yes
14	Number of new TEs added to roster	0.009	No
15	Normalized avg points per drive under HC	0.009	Yes
16	Normalized opponent rushing TDs under HC	0.009	Yes
17	Minimum percentage of games missed by RBs	0.009	No
18	Normalized passing yards under OC	0.009	Yes
19	Average experience of all starters	0.008	No
20	Net change in linebackers	0.008	No

D Win Percentage Persistence

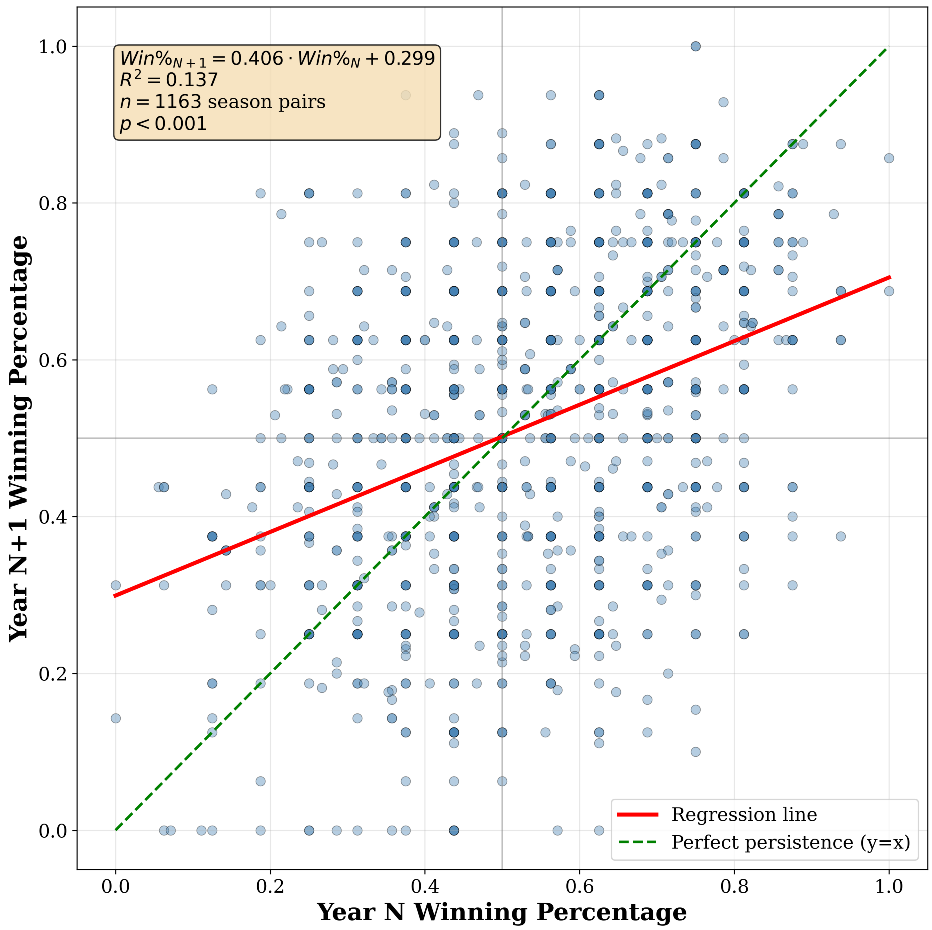

Fig. 7: Coach Win Percentage in Year N+1 vs. Year N

E WAR Persistence by Coach Background

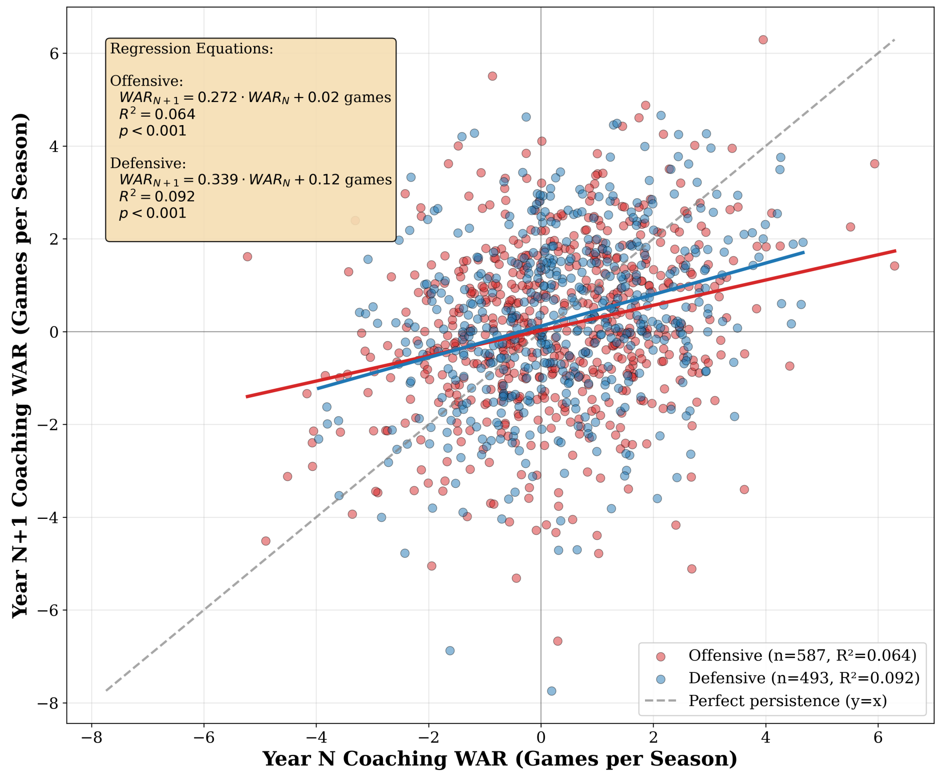

Fig. 8: Coach WAR in Year N+1 vs. Year N by Coach Background

Tab. 10: Coaching WAR Persistence by Background and Time Horizon

	Offensive		Defensive		Other	
Time Horizon	R^2	p-value	R^2	p-value	R^2	p-value
1-year	0.064	<0.001	0.092	<0.001	0.032	0.103
2-year avg	0.069	<0.001	0.180	<0.001	0.005	0.587
3-year avg	0.047	<0.001	0.153	<0.001	0.018	0.357
4-year avg	0.038	0.004	0.135	<0.001	0.019	0.429

Note: Fisher's Z-test for difference in correlations (Offensive vs Defensive): 1-year (p=0.380), 2-year (p=0.009), 3-year (p=0.020), 4-year (p=0.054).